

Appendix F: the per-stage probing subsection

A build plan. Everything here is writable now, costs no body words, and closes the one sub-part of lecturer comment A7 that is still open.

1. Why this is worth doing

Comment A7 asked four things of the MDL probing. Three are answered:

- *what representations will be probed* — Appendix B names ten states.
- *which metrics* — prequential description length in bits.
- *how results are interpreted against the hypotheses* — §6’s thresholds on θ_{lock} .

The fourth, ***how different decoder layers will be compared***, is not, and the current design actively prevents it. Model B carries stage as the nuisance offset s_{st} , deliberately absorbed so that level differences between probe points cannot leak into the estimand. That is correct for the estimand and useless for his question. Appendix F’s existing probe reading makes it worse: it uses **one** state, the output head.

So the answer has to be a descriptive per-stage reading, and an appendix is the only place it can live without spending body words. **The data already exist.**

2. The data you already have

`collect_md1_cells` writes one row per (probe stage, candidate frequency, geometry) into `o2_md1_cells.parquet`, with columns `model`, `P`, `S`, `stage`, `is_locked`, `L_bits`. The stage vocabulary is fixed in `probe_lib.py`:

Stage key	Depth	What it is
<code>enc_0 ... enc_3</code>	1–4	The [REG] token after each of the four encoder blocks.
<code>dec_0 ... dec_3</code>	5–8	The decoder state after each of the four decoder blocks. Named <i>decoder states</i> , not [REG] tokens — comment B1.
<code>output_reg</code>	9	The 256-dimensional pre-projection vector.
<code>output_head</code>	10	The projected quantile head. The only stage Appendix F reads today.

No new collection is required for the single-geometry reading. The cross-model layer needs `md1_cells` for the other geometries, which the same collection already produces.

3. The structure to build

Insert a new subsection into Appendix F **after** the decomposition-pages material and **before** the closing paragraph that begins “*None of this is evidence for or against the hypotheses*”. Placement matters: see §5.

F.x Where the frequency stops being readable

Paragraph 1 — what is being read and why it is here. State that Model B absorbs stage as a nuisance offset by design, so the fitted estimand says nothing about depth; that this subsection is the descriptive complement; and that it reports codelength by probe stage rather than pooled. ≈ 60 words.

Figure F.x1 — the depth profile, reference geometry. $P = S = 16$ retrained. x -axis: the ten stages in order, `enc_0` through `output_head`. y -axis: mean L_{bits} . **Two lines**, one for `is_locked=1` and one for `is_locked=0`, with a band for the spread across candidate frequencies. The figure’s whole purpose is the *gap between the two lines as a function of depth*: where it opens is where the frequency stops being linearly readable.

Paragraph 2 — reading the profile. ≈ 90 words. Three things the reader should be told to look for, whichever way the data fall:

- where the two lines separate — early (encoder, so the loss is at or near tokenisation), late (decoder or head, so the loss is in the readout), or nowhere;
- whether the absolute level rises or falls with depth, which is a property of the representation size and not of the phenomenon;
- that a gap at `output_head` alone would mean the projection, not the encoder, is where frequency identity is discarded — and that this is precisely the distinction the single-stage reading in the rest of the appendix cannot make.

Figure F.x2 — the same profile across geometries. Small multiples, one panel per geometry, same axes, same scale. Or, if space is short, one panel with the ten stages on x and one line per geometry showing the *locked-minus-control gap* rather than the raw level — the gap is comparable across geometries in a way the raw codelength is not. **Prefer the gap version**; it is the honest cross-model comparison. See caution 2.

Paragraph 3 — the cross-model reading and its limits. ≈ 90 words. Whether the depth at which the gap opens is stable across (P, S) or moves with the geometry; and the two caveats of §5 below, stated here rather than left implicit.

Table F.x — optional but cheap. Ten rows, one per stage; columns: mean L_{bits} locked, mean control, gap, and the number of cells. It makes the figure checkable and costs nothing.

4. What this closes

A7(c)	met. “How different decoder layers will be compared” is answered by a stated procedure and a figure that performs it: same probe task, same folds, same frequencies, compared across the ten stages, with the encoder and decoder halves distinguished.
A7(a)	Strengthened. The ten states are currently only <i>named</i> ; here they are <i>used</i> .
Requirement 6	Strengthened. “Representational evaluation” becomes a reading with internal structure rather than a single number.
Cost	Zero body words for the appendix. About 40 words in §9 Results, and those come <i>out of</i> the 600 budgeted there, not on top of it.

5. Four cautions, in order of how much damage they do

1. **Put it inside the descriptive firewall, not after it.** Appendix F ends on “*None of this is evidence for or against the hypotheses of §6*”. The new subsection goes *before* that sentence, and the sentence is extended to name it. Comment C9 requires H1 to rest on the amplitude-recovery and MDL *fits*, not on probe figures; a depth profile that reads as a verdict breaks C9 from the opposite direction. This is the caution that matters most, because the figure will be visually persuasive.
2. **Cross-geometry levels are not comparable raw.** Model B blocks on geometry with g_{geo} precisely because probe levels differ several-fold between configurations. Plot the locked-minus-control *gap*, and say in the text that the fitted model absorbs the level differences this figure leaves in.
3. **Say the comparison is like-for-like, and name what is not.** Same probe task, same folds, same candidate frequencies. Then state that the analysis context length already differs by geometry (512, 504, 500 for $S \in \{8, 16, 32\}, \{12, 24\}, \{20\}$). He raised exactly this about the sweep in comment C5; raising it yourself here, before he does, is worth more than the figure.
4. **Do not let the §9 sentence grow.** Two sentences: what the depth profile shows, and that it is descriptive. Forty words, inside the 600.

6. The sentence for §9 Results

Where §9 currently comments the sweep and the reconstruction over complex signals, add, in the same paragraph and in the same register:

The same appendix reads the probe by depth rather than pooled, so the stage at which a candidate frequency stops being linearly separable can be located within the encoder–decoder stack. Like the rest of that appendix it is descriptive, and the representational claim is decided by $e^{\theta_{\text{lock}}}$ under the rule of Table 5.

46 words. The second sentence is not padding: it is what keeps comment C9 satisfied while the first sentence points at a figure that looks like a result.

7. Order of work

1. Group `02_md1_cells.parquet` by `stage` and `is_locked` for `p16-s16`. If the two lines do not separate anywhere, that is still a finding and the subsection still answers A7(c) — write it either way.
2. Draw F.x1. Decide from it whether paragraph 2 says *early*, *late* or *nowhere*.
3. Draw F.x2 on the gap, across geometries.
4. Write the three paragraphs, insert before the firewall sentence, extend that sentence.
5. Add the 46-word sentence to §9, and subtract it from the 600.